Gate 1: Independent annual M1-LP certificate verification

Independent verification | 17 October 2026

1. Verdict and scope

CONDITIONAL PASS for all three annual separation statements. Three independent annual M1-LP restricted masters were reconstructed and solved, with no M2 optimization, no Stage II, no new debit family, no frozen runner and no historical file modified. The finite-box multipliers and two independently evaluated representations are retained.

The conclusions concern the fixed frozen numerical DC model. They are guarded floating-point certificates, with quantified forward-error diagnostics and independently admissible multiplier sums. They are not fully outward-rounded interval certificates. The archived M2 upper endpoints remain numerical replay values; an additional explicit replay-error allowance is reported.

2. Authoritative inputs and ordering

The explicit 17 October 2026 main and companion attachments are byte-identical to the latest audit-integrated ZIP; AUTHORITY.json records their complete SHA-256 identities. The physical parent case is 2ed58c3ba8dd673cc53996f713acb983126a9df45922afcb2d5adb3c2e34aeb6; the physical arrays SHA-256 is a778a4c2bc2645969cf7764ff10dc88fa114ebd0c9e295edcc25aaaf07c89feb.

Annual child identities, time-grid hashes, position order, masks, annual allowance hashes, 94 state hashes, 8,837 row identities, and archived geometry hashes were independently reconstructed. Every year has 8,760 one-hour intervals and 114 directional positions. The 45-bus counter-injection matrix and signed state PTDF rows are read directly from the immutable physical arrays.

Year	Annual physical hash / geometry hash
2029	fecda0c0d1996287ab4636a807d007459ab5670367f59406803c0a0c764df707 / 1f23c01b01e95cec9d74dcd97c9f905944dbe56b8056e53507d722260905dc00
2030	6651f1de950cd71fdf80967d93664a9e86b09ed8eb884d92b190e604d70232d6 / d61258ee1bec81c76e60a4cbef6986d8e05f3c1aa04cb3cd3c9b341a92a7ec4a
2031	94c0bbb39c2c050f8c6de1678bb01cf4f21dfe07b4962d90a257557bf2744073 / db7b9087a59bd5792da3120625778f53b94ae346c7576b261c4b7c6007101103

3. Mathematical problem

$b_{nt} = \chi_{nt} \min(P_n, v_{nt})$, $0 \le x_{nt} \le b_{nt}$. The annual M1-LP ledger is $\Sigma_{t:b_{nt}>0} \Delta_t x_{nt}/b_{nt} \le H_n$, with $H_n = \text{floor}(0.15 \Sigma_t \text{contract_mask}_{nt} + 10^{-12})$. The denominator is the eligible requested bound b_{nt}, not P_n. Zero-bound coordinates are substituted as exactly zero.

$p_{req} = p_{background} - A v$; $f_{req} = H_{network} p_{req} + h$; $G = H_{network} A$. The complete primary objective is $S(x) = \Sigma_t \Delta_t \Sigma_j [|f_{req,tj} + G_j x_t| - F_j]_+$. Every signed monitored state-element hinge is retained in this mathematical objective. No weighted cost, redispatch objective or Stage-II regularizer is substituted.

The verification master has x for every positive b, $\theta_t \ge 0$ for all 8,760 hours, budget $Dx \le H$ and selected supports $a_k x_t - \theta_t \le -\beta_k$. It minimizes $\Sigma_t \theta_t$. Here $C0 = 0$ exactly, and no hour is also counted in an external constant.

4. Screening and feasible-domain proof

The historical geometry and retained-coordinate order are reproduced exactly, including safe-row selectors, potential positive/negative hinges and original objective constant. However, the verification does not impose historical useful-coordinate screening or secure-hour removal. It uses the full positive physical action box. This is an explicitly authorized valid relaxation, not a claim that the larger restricted master has identical dimensions to the historical screened master.

For any complete-model action x, set each θ_t equal to its complete hourly hinge sum. Every selected support is then satisfied. Thus every original feasible action lifts into the new master, and the new master optimum is a lower bound on the original complete M1-LP optimum. Omitted hinges need not be proven inactive: they have zero dual weight in this relaxation. The historical inactive-row test was nominally reproduced, but its floating-point threshold is not used as a rigorous interval screening theorem.

The historical dominated-coordinate reasoning is also sound at the exact mathematical level if setting that coordinate to zero decreases every potentially active signed hinge and decreases a nonnegative debit. Requested-secure hours can then be set to zero at an optimum. These observations explain the old reductions but are not assumptions of the new certificate.

5. Supporting-plane verification and coefficient errors

For each archived selector k, its positive and negative row lists are unique, belong to the registered annual active hour and registered potential-row sets, and preserve state-element and sign identities. A selected directed hinge contributes G_j and f_req_tj−F_j for positive orientation, and −G_j and −f_req_tj−F_j for negative orientation. Summing any subset gives an affine global minorant because [z]_+ ≥ z for selected hinges and [z]_+ ≥ 0 for omitted hinges. Durations equal one hour in these annual problems.

Year	Supports / directed selectors	Max slope mismatch	Max intercept mismatch	Max support-formation error estimate
2029	82685 / 21327283	0.0	2.91e-11	4.66401e-07
2030	74297 / 8276722	0.0	5.82e-11	4.95559e-07
2031	112851 / 7270126	0.0	2.91e-11	5.18942e-07

The source-compatible slope sums are checked against an independently assembled full-position sum. Intercepts are compared with math.fsum. Small positive support-minus-complete replay differences arise from floating arithmetic and are covered by the reported plane-formation allowance. Standard γ_k = ku/(1−ku), u = 2^-53, is used for absolute dot/sum formation estimates, including H A and requested-flow formation. These estimates do not enclose uncertain historical real-world data and are not outward interval arithmetic.

6. LP dimensions and solver record

Year	Full-box x / θ	Rows / nonzeros	Historical x
2029	354780 / 8760	82799 / 4561832	282048
2030	411720 / 8760	74411 / 4744548	331272
2031	459900 / 8760	112965 / 7821981	418948

Gurobi 12.0.3; one thread; dual simplex Method=1; Presolve=2; FeasibilityTol=OptimalityTol=10^-9; NumericFocus=2; Seed=0; TimeLimit=3600 seconds; default solver scaling and original physical/accounting units. These independent-verification settings differ from the historical run and do not change its mathematical coefficients. Each annual model has one optimize call and no added supports. MPS, basis, settings, log, primal vector, raw row duals, reduced costs and basis statuses are retained.

Year	Status	Primal objective	Solver bound	Runtime s	Iterations
2029	OPTIMAL	57909827.751592740	57909827.751592740	120.11	121549.0
2030	OPTIMAL	37782287.252002701	37782287.252002701	72.80	97300.0
2031	OPTIMAL	23658821.385054886	23658821.385054886	455.40	249752.0

7. Independent finite-box dual derivation

The ≤ budget and support rows have nonpositive Gurobi Pi values. Use λ = max(−Pi_budget,0), ζ = max(−Pi_support,0). Scale each near-saturated hour toward 1−2^-35; then check the actual stored binary multipliers by exact rational summation to verify Σ_{k at t} ζ_k ≤ 1. All 114 budget multipliers and every support multiplier are saved.

The Lagrangian is C0 − H^Tλ + Σ_k ζ_k β_k + r^Tx + Σ_t(1−Σ_{k at t}ζ_k)θ_t, where r = D^Tλ + $A_{support}^T$ζ. Its infimum over θ≥0 and 0≤x≤b is L0 = C0 − H^Tλ + Σ_k ζ_k β_k + Σ_j b_j min(r_j,0). This proves a lower bound for any admissible multipliers; exact stationarity and master optimality are unnecessary.

The verifier uses no solver import. It reconstructs r via sparse support columns, and independently aggregates ζ onto the original positive and negative physical rows at each hour before applying G. The two intercept and box-correction evaluations are compared. This is stronger evidence than copying ObjVal or ObjBound.

8. Complete bound components and numerical guard

Year	−H^Tλ	Σ ζ β	Finite-box correction	Downward guard	Guarded new L
2029	-64107495.358244650	124341141.455282107	-2323818.347685257	9.41634221e-05	57909827.749258034
2030	-25497630.803920589	76369602.961739153	-13089684.907249665	2.92326475e-05	37782287.250539653
2031	-22434405.472943578	47034595.383941598	-941368.527215995	9.67953391e-06	23658821.383772343

The downward guard sums: (i) ζ-weighted uniform support-formation/error estimates on the full box; (ii) b-weighted sparse accumulation and budget-quotient uncertainty in r; and (iii) scalar product/reduction error. A final upward rounding is applied to the guard and downward rounding to L0−guard. The guard is derived from coefficient magnitudes and operation counts, not selected to obtain the desired separation.

This implementation does not outward-round every arithmetic step. Its formal exact-arithmetic theorem is separate from the guarded floating numerical realization. The machine-checkable archive retains all components and the unguarded value for independent higher-precision or interval work.

9. Numerical residuals and independent-path agreement

Year	Budget violation h	Support violation	Box violation MW	Row-path − support-path	Guarded primal-dual gap
2029	2.046e-12	5.402e-10	0	0	0.00233471
2030	1.364e-12	3.311e-10	8.882e-16	7.45058e-09	0.00146305
2031	1.251e-11	6.366e-11	0	0	0.00128254

The complete residual CSV also records raw dual reconstruction, exact hourly multiplier violations (zero), stationarity-path discrepancy, replay differences and the margin after upper-endpoint error. Solver quality attributes, Kappa/KappaExact where available and native MaxMemUsed are in SOLVER_RESULT.json. Gurobi reports native memory in its own GB units; this is distinct from historical process memory measured in GiB.

10. Original M2 witness verification

No annual M2 optimization or new repair is performed. Each original selected solution array is checked against its stored SHA-256. Multiplying it by the already archived B03 positionwise scaling factors reproduces the archived B03 witness hash. Every stored witness coordinate satisfies 0≤x≤b, and Σ_t x_nt/P_n≤H_n is checked using exact rational representations of the stored binary coordinates and capacities. Δ_t=1 for these years.

The raw selected M2 policy and the B03 witness are different numerical objects. Only the archived scaled witness is used as the upper endpoint. Its full 8,837-row, 8,760-hour thermal residual is replayed through p_req+A x followed by H, independently of the cached base+Gx path. Units are MW·h summed over monitored state-element rows.

Year	Archived witness U	Full replay − U	Replay error estimate	Exact box / ledger
2029	48058421.890843324	-2.23517418e-08	0.00604650998	PASS / PASS
2030	32237842.052070111	-7.4505806e-09	0.0040554235	PASS / PASS
2031	18740819.467404809	-3.7252903e-09	0.00235423815	PASS / PASS

The replay error allowance includes forward-error estimates of request and action injection formation, state-row evaluation and reductions. U remains the original archived numerical endpoint; the stricter margin L−U−δ−replay_error is also positive. This does not convert U into a fully interval-certified endpoint.

11. Strict separation and historical reconciliation

δ_y = max(10^-6,10^-9 max(1,|U_y|,|L_y|)). The numerical separation is U_y+δ_y<L_y. The reported reduction is 100[1−(U_y+δ_y)/L_y], with the new L in the denominator.

Year	Historical L	New L	δ	L−U−δ	Reduction %
2029	57909827.751589	57909827.749258	0.057909828	9851405.800505	17.011630
2030	37782287.252000	37782287.250540	0.037782287	5544445.160687	14.674721
2031	23658821.385053	23658821.383772	0.023658821	4918001.892709	20.787180

The conclusions do not require equality to the historical lower-bound scalars. Differences reflect an independently assembled complete-box restricted master, independently projected duals and a new quantified numerical guard. The sign-off manuscript must use these newly retained bounds and explicitly preserve their finite-precision qualification.

12. Reproduction, file ordering and remaining condition

GATE1_VERIFICATION_CODE contains standalone Python code. Dependencies: Python 3.9+ with NumPy, SciPy, pandas and Gurobi for the three solves. prepare_gate1.py reconstructs all models and writes PRE_SOLVE_GATE.json before solving. solve_gate1.py has exclusive per-year OPTIMIZATION_STARTED.json markers to prevent accidental repeats. verify_certificate.py YEAR is solver-free and can be rerun on retained artifacts; no original experiment runner is invoked.

For a new verification, copy the scripts to a fresh isolated sign-off layout and deliberately configure its paths. Do not rerun preparation in this sealed completed directory. The solve cap applies to this task, not a standing authorization for future recomputation. The full matrices and vectors remain private evidence outside the publisher upload ZIP.

Ordering: x coordinates are (variable_hours, variable_positions) in row-major positive-box order; θ follows all 8760 year hours; support multipliers follow archived cut order; hour mapping is explicit; row selectors use the exported NETWORK_ROW_MAPPING.csv. Aggregated row-dual CSR matrices have shape8760×8837. Full raw solver vectors and projected admissible vectors are separate lossless NPZ files.

Remaining numerical condition: standard guarded floating-point evidence is strong and the strict margins are large, but a complete outward-rounded interval proof has not been produced. Gate 3 may proceed with explicit qualification. Original scientific-source integrity is independently checked before final packaging.